DTFT:

Properties:

$$x[n] * h[n] \longleftrightarrow X(\Omega)H(\Omega)$$

$$y[n]=x_1[n]\,x_2[n] \longleftrightarrow \underbrace{\frac{1}{2\pi}\int_0^{2\pi} X_1(\tau)X_2(\Omega-\tau)\,d\tau}_{\text{Periodic Convol}^n} = Y(\Omega)$$

$$\underbrace{x[n] \longrightarrow \boxed{\text{DTFT}} \longrightarrow X(\Omega)}_{\text{Linear \& invertible}}$$

Pf:

$$\frac{1}{2\pi}\int_0^{2\pi} Y(\Omega)e^{j\Omega n}\,d\Omega$$

$$= \frac{1}{2\pi}\int_0^{2\pi}\frac{1}{2\pi}\int_0^{2\pi} X_1(\tau)X_2(\Omega-\tau)\,d\tau\; e^{j\Omega n}\,d\Omega$$

$$= \frac{1}{2\pi}\int X_1(\tau)\,d\tau\,\underbrace{\frac{1}{2\pi}\int_0^{2\pi} X_2(\Omega-\tau)e^{j\Omega n}\,d\Omega}_{= x_2[n]e^{j\tau n}\ (\text{Inverse DTFT of } X_2(\Omega-\tau))}$$

$$= \frac{1}{2\pi}\int_0^{2\pi} X_1(\tau)x_2[n]e^{j\tau n}\,d\tau$$

$$= \frac{1}{2\pi}x_2[n]\int_0^{2\pi} X_1(\tau)e^{j\tau n}\,d\tau$$

$$y[n] = x_1[n]\,x_2[n]$$

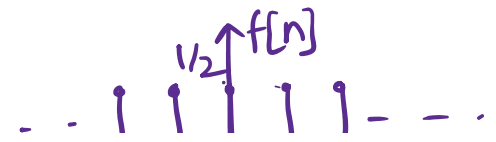
$$\underbrace{\sum_{k=-\infty}^{n} x[k]}_{x[n]*u[n]\ (\text{Accumulator})} \longleftrightarrow$$

$$x[n]*u[n]\ (\text{Accumulator}) \longleftrightarrow X(\Omega)U(\Omega)$$

$$U(\Omega) = \sum_{n=-\infty}^{\infty} u[n]e^{-j\Omega n} = \sum_{n=0}^{\infty} e^{-j\Omega n} = \underbrace{\sum_{n=0}^{\infty}\alpha^n = \frac{1}{1-|\alpha|}}_{\text{Not converging}},\ |\alpha|<1$$

$$u[n] = f[n] + g[n]$$

$f[n] = \frac{1}{2}\forall n$



$$\Big\} \longrightarrow \pi\sum^{\infty}\delta(\Omega-2\pi\ell)$$

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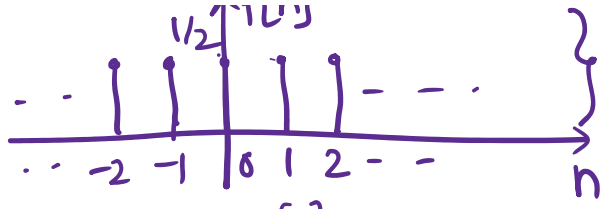
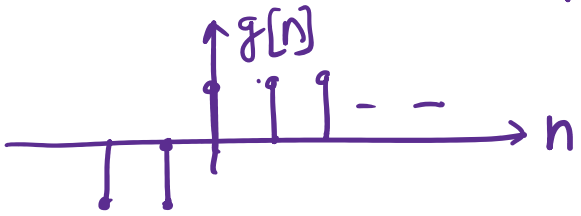
$f[n] = \frac{1}{2} \; \forall n$



$\longrightarrow \pi \sum_{l=-\infty}^{\infty} \delta(\Omega - 2\pi l)$

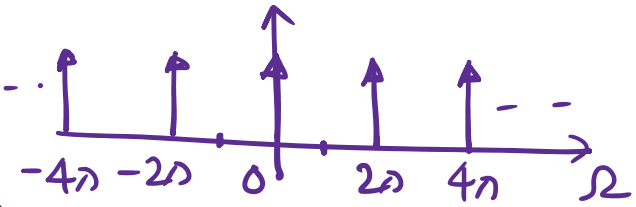
$g[n] = \begin{cases} \frac{1}{2}, & n \geq 0 \\ \frac{-1}{2}, & n < 0 \end{cases}$



$g[n] - g[n-1] = \delta[n]$

$\delta[n] \longleftrightarrow 1$

$1 \longleftrightarrow 2\pi \sum_{l=-\infty}^{\infty} \delta(\Omega - 2\pi l)$



<u>Pf:</u> $\frac{1}{2\pi} \int_{-\pi}^{\pi} \sum_{l=-\infty}^{\infty} 2\pi \, \delta(\Omega - 2\pi l) e^{j\Omega n} d\Omega$

$$\int_{-\pi}^{\pi} \delta(\Omega) e^{j\Omega n} d\Omega = \int_{-\pi}^{\pi} \delta(\Omega) \cdot 1 \, d\Omega = 1$$

$$f[n] = \frac{1}{2} \longleftrightarrow \pi \sum_{l=-\infty}^{\infty} \delta(\Omega - 2\pi l) = F(\Omega)$$

$$g[n] - g[n-1] = \delta[n] \longleftrightarrow G(\Omega) - G(\Omega) e^{-j\Omega} = 1$$

$$\Rightarrow G(\Omega)\left[1 - e^{-j\Omega}\right] = 1$$

$$\Rightarrow G(\Omega) = \frac{1}{1 - e^{-j\Omega}}$$

$$\therefore U(\Omega) = F(\Omega) + G(\Omega) = \frac{1}{1 - e^{-j\Omega}} + \pi \sum_{l=-\infty}^{\infty} \delta(\Omega - 2\pi l)$$

$$\therefore \sum_{k=-\infty}^{n} x[k] \longleftrightarrow X(\Omega) U(\Omega)$$

$$= \frac{X(\Omega)}{1 - e^{-j\Omega}} + \pi X(0) \sum_{l=-\infty}^{\infty} \delta(\Omega - 2\pi l)$$

$x[n] \longrightarrow \boxed{h[n]} \longrightarrow y[n]$

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Let $x[n] = \alpha^n u[n]$, $h[n] = \beta^n u[n]$, $|\alpha| < 1$ $|\beta| < 1$

$$\Rightarrow y[n] = x[n] * h[n]$$

$$\Rightarrow Y(\Omega) = X(\Omega) H(\Omega)$$

$$= \frac{1}{1-\alpha e^{-j\Omega}} \cdot \frac{1}{1-\beta e^{-j\Omega}} = \frac{A}{1-\alpha e^{-j\Omega}} + \frac{B}{1-\beta e^{-j\Omega}}$$

$$= \frac{A(1-\beta e^{-j\Omega}) + B(1-\alpha e^{-j\Omega})}{(1-\alpha e^{-j\Omega})(1-\beta e^{-j\Omega})}$$

$$\Rightarrow A = \frac{\alpha}{\alpha-\beta} , B = \frac{-\beta}{\alpha-\beta}$$

$$\Rightarrow Y(\Omega) = \frac{1}{\alpha-\beta}\left[\frac{\alpha}{1-\alpha e^{-j\Omega}} - \frac{\beta}{1-\beta e^{-j\Omega}}\right]$$

$$\Rightarrow y[n] = \frac{1}{\alpha-\beta}\left[\alpha^{n+1} u[n] - \beta^{n+1} u[n]\right]$$

$$x[n] \longrightarrow \boxed{h_1[n]} \longrightarrow \overset{\text{Inverse}}{\boxed{h_2[n]}} \longrightarrow x[n]$$

$$h_1[n] * h_2[n] = \delta[n]$$

$$\Rightarrow H_1(\Omega) H_2(\Omega) = 1 \Rightarrow H_2(\Omega) = \frac{1}{H_1(\Omega)}$$

$$h_1[n] = \left(\frac{1}{5}\right)^n u[n] , h_1[n] - A h_1[n-1] = \delta[n]$$

$$\Rightarrow H_1(\Omega) = \frac{1}{1-\frac{1}{5}e^{-j\Omega}} \qquad \Rightarrow A = 1/5$$

$$\Rightarrow h_2[n] = \delta[n] - \frac{1}{5}\delta[n-1]$$

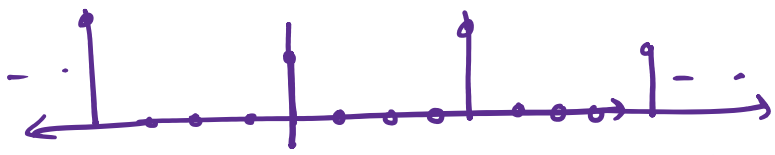
$$\Rightarrow H_2(\Omega) = 1 - \frac{1}{5}e^{-j\Omega}$$

$$\Rightarrow h_2[n] = \delta[n] - \frac{1}{5}\delta[n-1]$$

$$x[n] \longrightarrow \boxed{\overset{\text{LTI}}{h[n]}} \longrightarrow y[n]$$

$$X(\Omega) \longrightarrow \underbrace{H(\Omega)}_{\text{Freq. Response}} \longrightarrow Y(\Omega) = X(\Omega)H(\Omega)$$

Ex: $x_k[n] = \begin{cases} x[n/k], & n = 0, \pm k, \pm 2k, \ldots \\ 0, & \text{o.w.} \end{cases}$



} Expansⁿ in time domain

$$X_k(\Omega) = \sum_{n=-\infty}^{\infty} x_k[n] e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega nk} = X(k\Omega)$$

} Compresⁿ in freq. domain

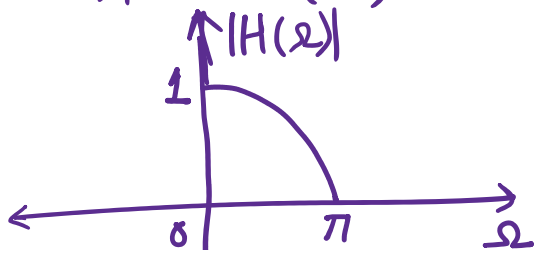
∴ Signal can't be localized in time domain & freq. domain (to arbitrary precision). } Uncertainity principle.

$$\frac{x[n] + x[n-1]}{2} = y[n]$$

$$Y(\Omega) = \frac{1}{2} X(\Omega)\left[1 + e^{-j\Omega}\right] = X(\Omega)H(\Omega), \quad H(\Omega) = \frac{1 + e^{-j\Omega}}{2}$$

$$H(\Omega) = \frac{e^{-j\frac{\Omega}{2}}}{2}\left[e^{j\Omega/2} + e^{-j\Omega/2}\right] = e^{-j\Omega/2}\cos\left(\frac{\Omega}{2}\right)$$

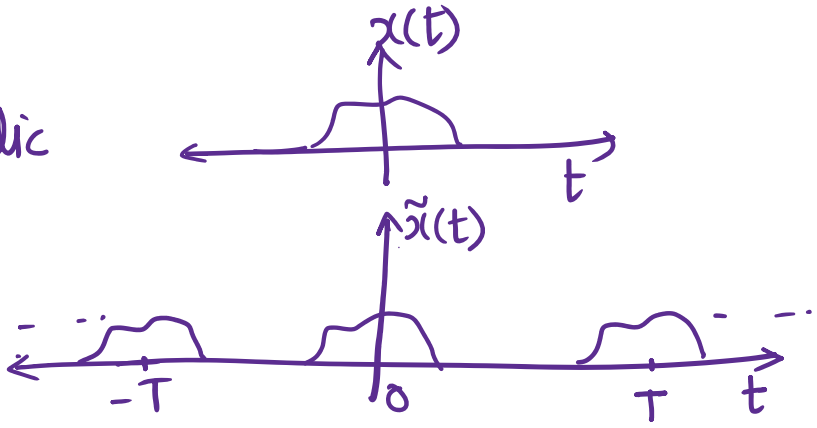
$$|H(\Omega)| = \cos\left(\frac{\Omega}{2}\right)$$



} Simplest LPF

CTFT:

Let $x(t)$ be aperiodic



$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} x(t-kT) \longrightarrow \text{Periodic with } T$$

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$a_k = \frac{1}{T}\int_0^T \tilde{x}(t) e^{-jk\omega_0 t}\,dt = \frac{1}{T}\int_0^T x(t) e^{-jk\omega_0 t}\,dt = \frac{1}{T}\int_{-\infty}^{\infty} x(t) e^{-jk\omega_0 t}\,dt$$

$$X(\omega) \triangleq \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\,dt$$

$$\omega_0 = \frac{2\pi}{T}$$

$$\Rightarrow a_k = \frac{1}{T} X(k\omega_0)$$

$$\Rightarrow \tilde{x}(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T} X(k\omega_0) e^{jk\omega_0 t} = \frac{1}{\cancel{T}} \sum_{k=-\infty}^{\infty} X(k\omega_0) e^{jk\omega_0 t} \frac{2\pi}{T} \cdot \frac{\cancel{T}}{2\pi}$$

$$= \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(k\omega_0) e^{jk\omega_0 t} \omega_0$$

Let $T \longrightarrow \infty \Rightarrow \tilde{x}(t) = x(t)$ & $\omega_0 \longrightarrow 0$

$$\Rightarrow x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t}\,d\omega \quad \Big\} \text{ CTFT Synthesis}$$

$$\& \; X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\,dt \quad \Big\} \text{ CTFT Analysis}$$